Assessing the Impact of Dataset Characteristics —Homogeneity Composition and Data Sparsity— on AI Model Transparency, Fairness, and Bias Evaluation

Examining Bias and Unfairness in Explainable Artificial Intelligence (XAI) Due to Imbalanced Training Data and Misleading Effects of Small Sample Sizes

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Abstract

Addressing the **Characteristics of the Dataset**, including its homogeneity composition and Data Sparsity, is critical for a comprehensive approach to Responsible AI (XAI).

1. **Homogeneity Composition**: balanced vs imbalanced datasets.

2. **Data Sparsity**: small vs large datasets.

Agenda

- Introduction
- Literature Review
- Methodology
- Result Findings
- Discussions
- Conclusions
- Bibliography

01. Introduction

This study proposes the incorporation of dataset characteristics to be documented into Responsible AI and Explainable AI (XAI) practices,
especially the **homogeneity composition**, whether the dataset is balanced or imbalanced,
and **data sparsity**, whether small or large samples,
for transparency at both the data and model levels

Research Questions

- 1. How does the dataset homogeneity composition (balance vs. imbalanced) impact the AI model transparency and fairness in decision-making systems?
- 2. How does the data sparsity/ dataset's size (Large versus small) influence the accuracy of bias evaluation in AI models, when the training data is class-imbalanced?

02. Literature Review

2.1. Overview of Relevant Previous Study

Based on research studied by Miao, Y., Cao, H., and Tang, M. (2022), the **imbalance ratio (IR) influences the performance and stability of the model.** The models like Logistic Regression, Decision Trees, and Support Vector Classifiers show less stability, while Naive Bayes, Random Forests, and Gradient Boosting Decision Trees showed more stable performance.

The study by Lin (2018) with title "Bias caused by sampling error in meta-analysis with small sample sizes" warned that **too small sample sizes can create significant bias**.

2.2. Terminology and Definition

ACCURACY SHOWS THE OVERALL CORRECTNESS OF THE MODEL.

PRECISION IS A METRIC THAT MEASURES HOW OFTEN A PREDICTED LABEL IS CORRECT.

RECALL IS A METRIC THAT MEASURES HOW OFTEN A TRUE LABEL IS CORRECTLY PREDICTED.

F1-SCORE IS THE HARMONIC MEAN OF PRECISION AND RECALL.

AUC-PR IS PRECISION-RECALL CURVE**,** SHOWS OVERALL PERFORMANCE OF THE MODEL.

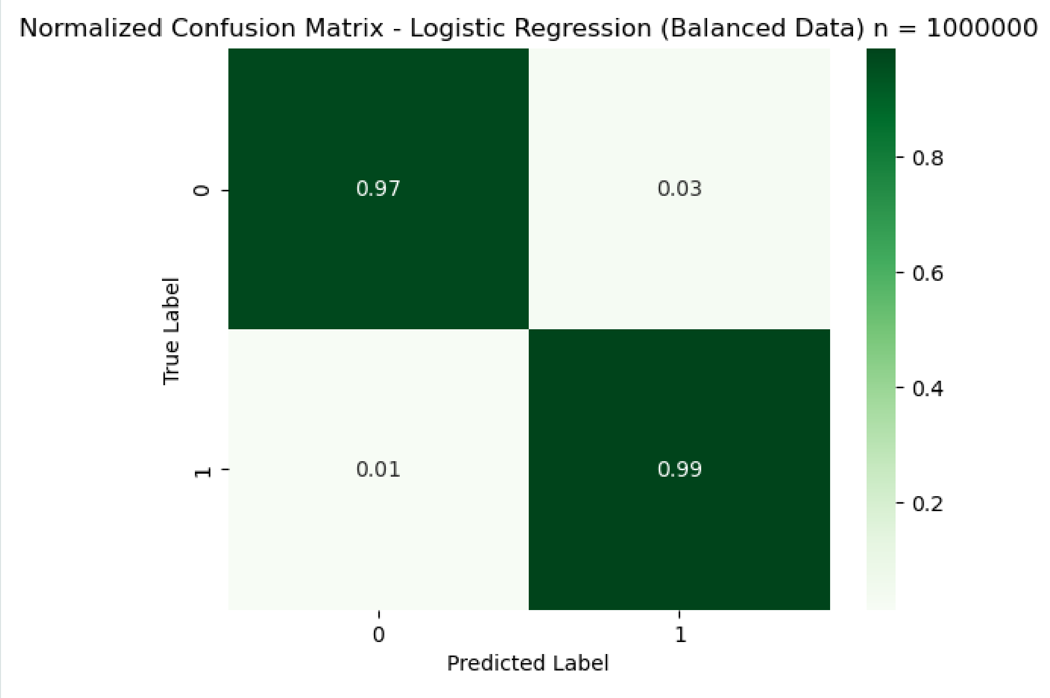
A HIGHER AUC VALUE INDICATES BETTER PERFORMANCE.

Methodology

- Synthetic/ Artificial Datasets
- **Small Balanced** Datasets (n = 10000) vs **Large Balanced** Datasets (n = 1000000)
- **Small Imbalanced** Datasets (n = 10000) vs **Large Imbalanced** Datasets (n = 1000000)
- Logistic Regression
- LIME (local interpretable model-agnostic explanations)

04. Result Findings

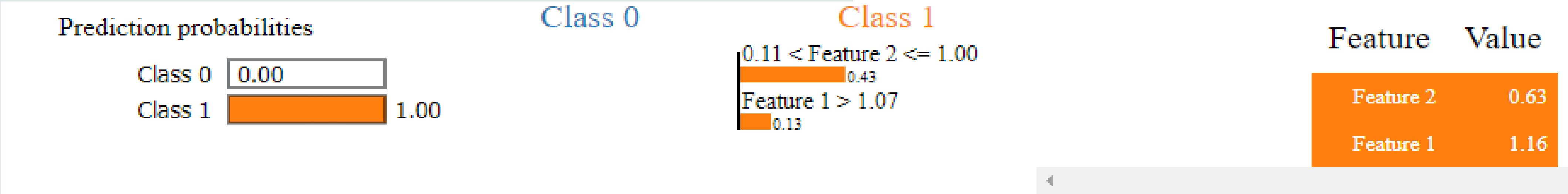
Large and Class-Balanced Dataset



Good for all classes.

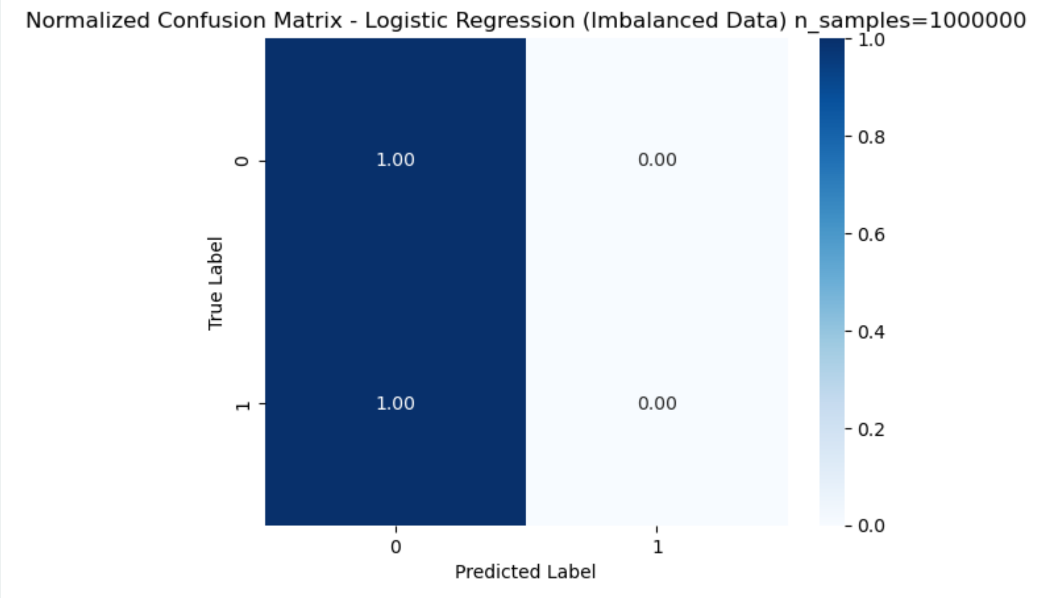
Both features contribute to both classes.

No class was ignored.




```
# Alternatively, to print the explanation in the console
print(explanation_bal.as_list())
```

```
[('0.11 < Feature 2 <= 1.00', 0.43234393819941574), ('Feature 1 > 1.07', 0.13107239803042026)]
```

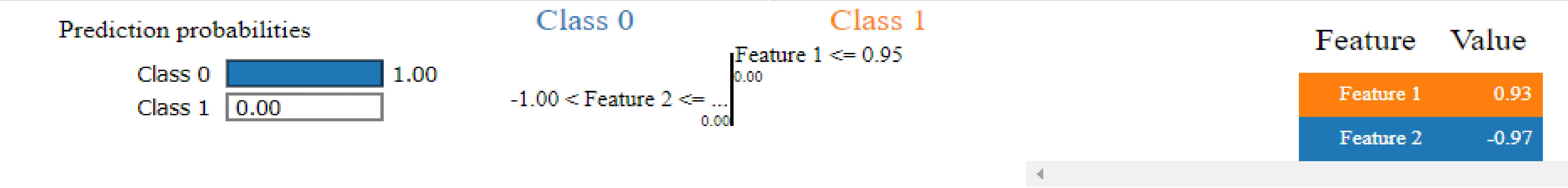


Large and Class-Imbalanced Dataset

There is a **bias towards the majority class** in Class-Imbalanced Dataset.

The model predicts **Class 0** with 100% confidence.
Probability of Class 1 is zero.

Class 1 is the minority class, and the model learned to totally ignores the probability of class 1 completely.

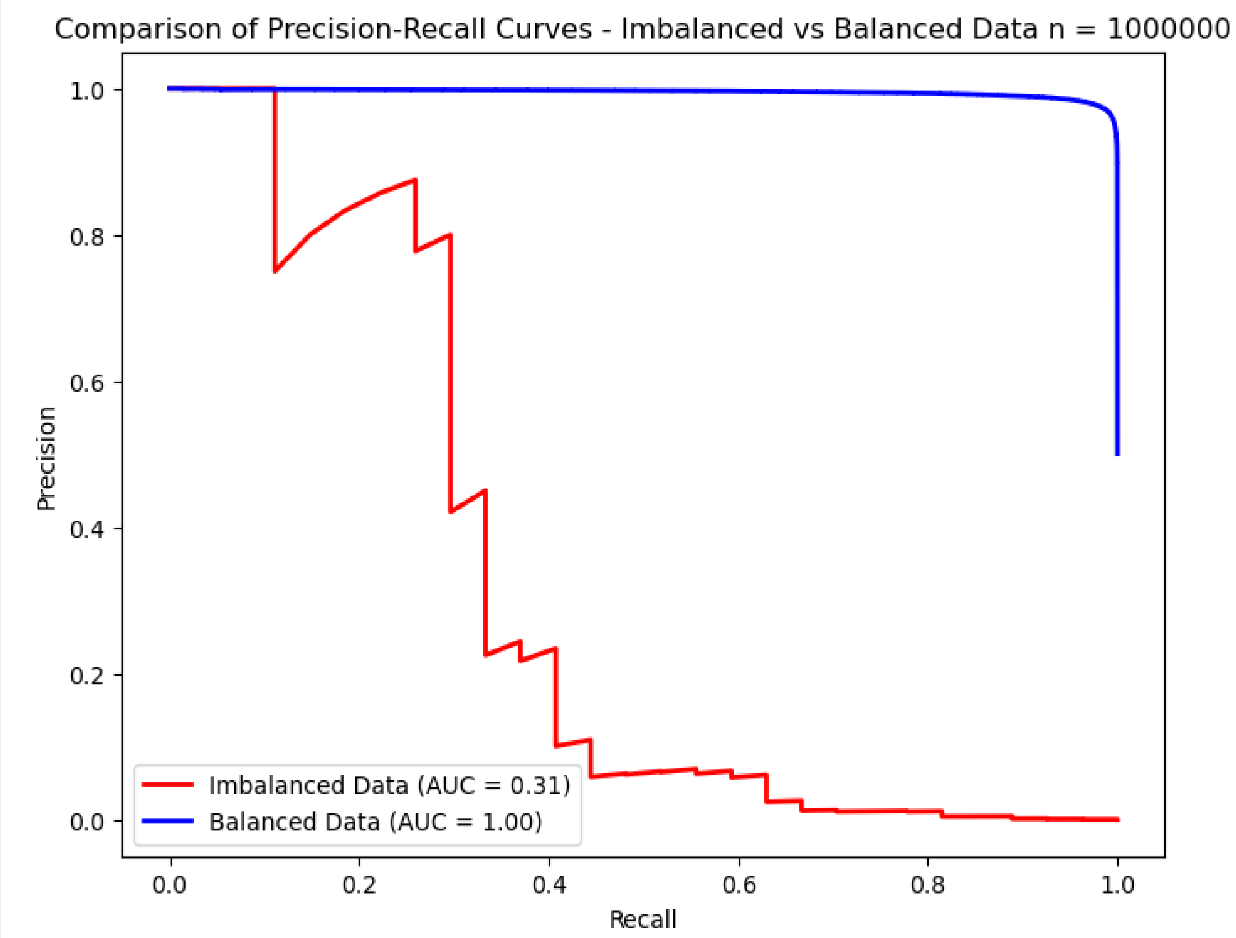



```
print(explanation.as_list())
```

```
[('Feature 1 <= 0.95', 7.824398545718031e-05), ('-1.00 < Feature 2 <= -0.54', -6.006037438071732e-05)]
```

Large Size Dataset Comparison

A low AUC value means that the prediction for the class is poor and has bias. A high AUC value means that the prediction for the class is good and has no bias.



The area under the curve (AUC) for Class-Balanced Data is higher than Class-Imbalanced Data.

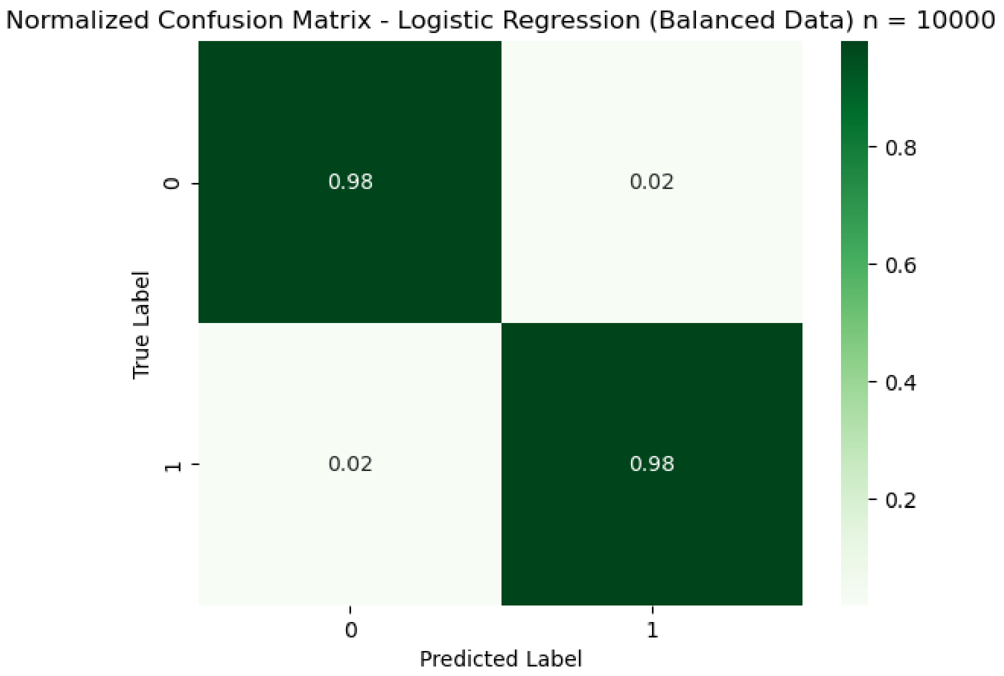
The model performs very well when the dataset is balanced.

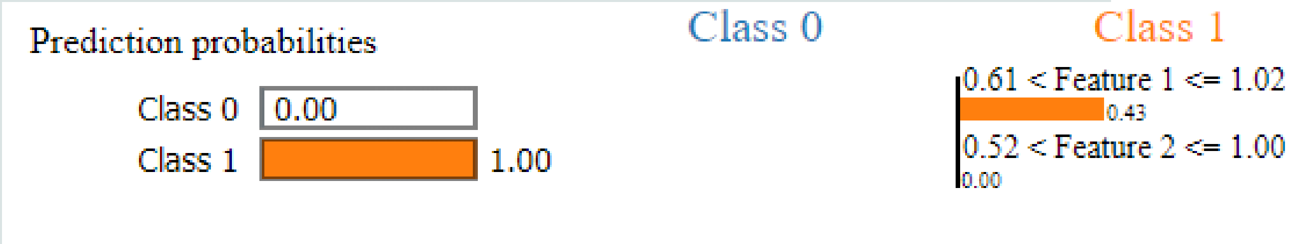
The model's true inability to predict the minority class becomes obvious, with a significant drop in AUC from 0.97 to 0.31.

Small and Class-Balanced Dataset

The model can considering all features.

Both features contribute to all classes. No class was ignored.



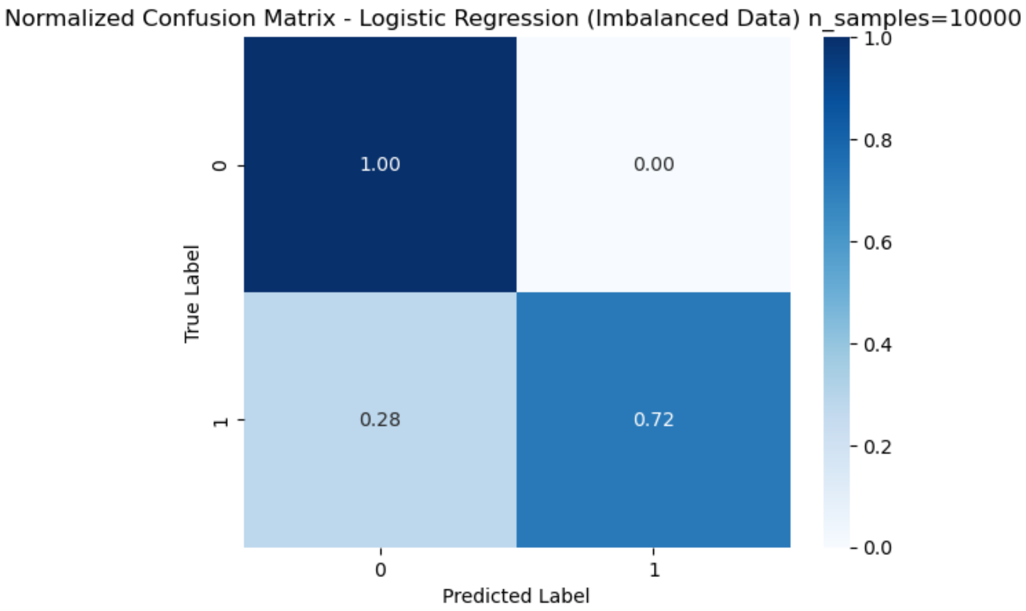


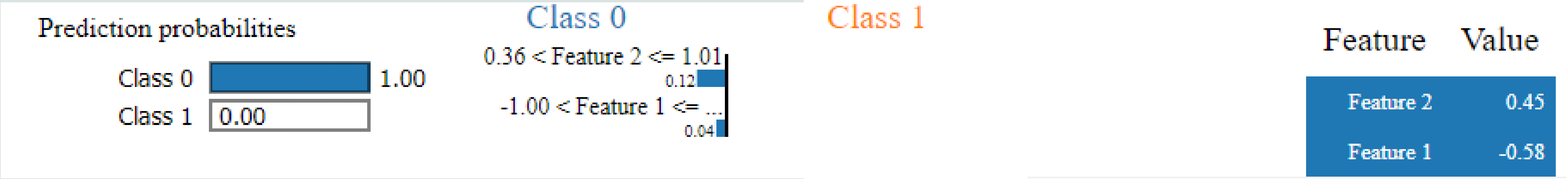
Feature	Value
Feature 1	0.84
Feature 2	0.70

Small and Class-Imbalanced Dataset

There is a bias in class-imbalanced dataset. But the **small sample size can’t detect it**.

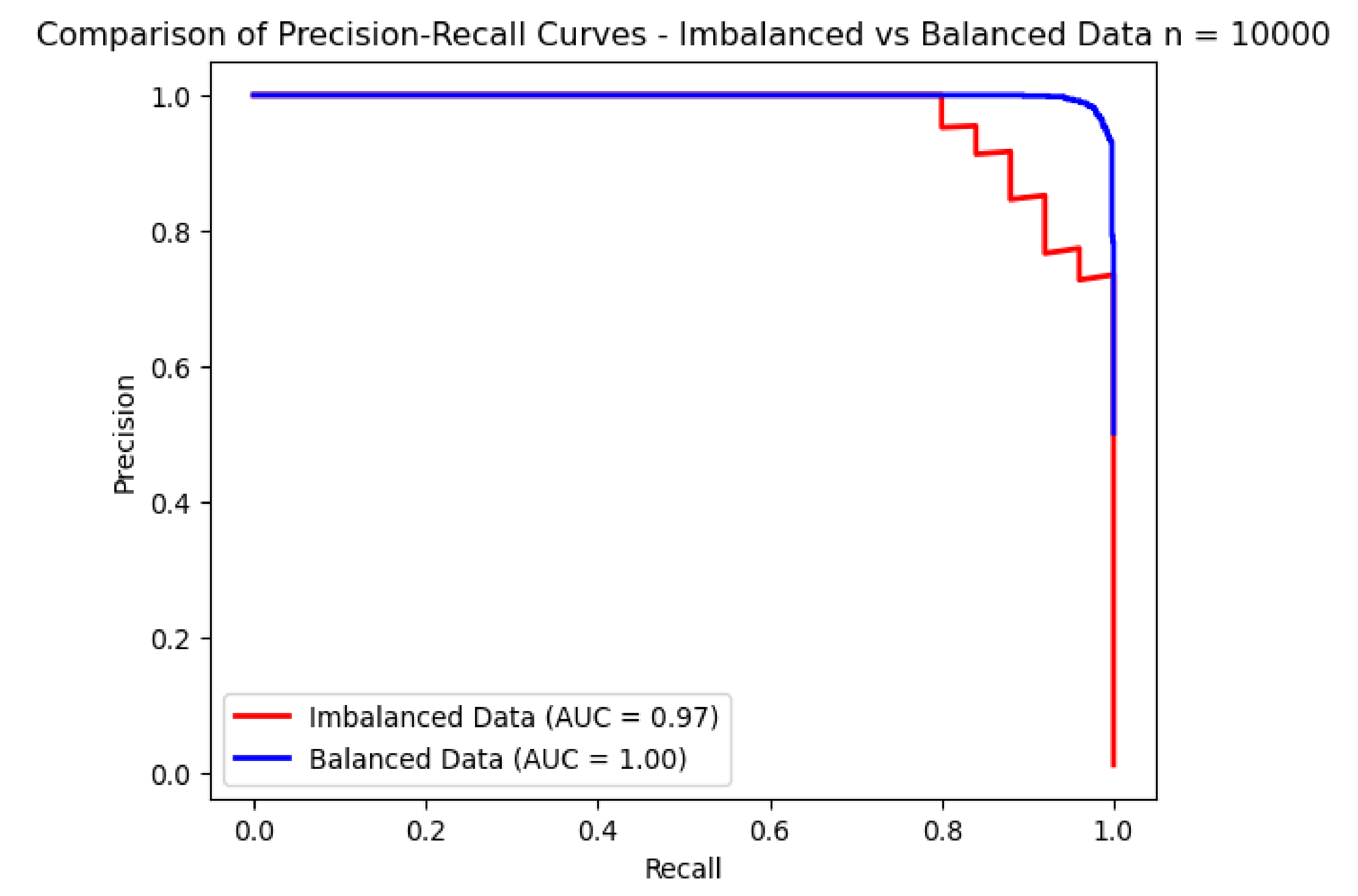
In small sample size, the performance metrics **fakes** the bias on class-imbalanced dataset.





Small sample size Dataset Comparison

Smaller sample sizes can fake the bias of class imbalance, making fake perception of model fairness.



Result

There is **bias in class-imbalance dataset**. But small sample size can't detect it.

Large data set can show the true face of the bias, provides more accurate and reliable evaluation of models.

In **small and imbalanced datasets**, model often shows a **fake** high performances metrics, gives **misleading results.**

Dataset Characteristic	AUC
Small Balanced	1.00
Small Imbalanced	0.97
Large Balanced	1.00
Large Imbalanced	0.31

05. Discussions

A significant impact of dataset composition on model fairness and transparency

Class-Imbalanced data set results in **bias** in the model that discriminatively and unfairly predicts directly for the majority class. Minority groups are not represented.

Small Sample Size gave misleading impression of fairness, because it can fake the model's performance on minority classes, when in fact it does not.

Bias Evaluation is failed when using small class-imbalanced data sets

Discussion of Limitation

Several limitations of this research

This study only evaluates **two** types of data characteristics, which are only data **homogeneity** and **data sparsity**

This study only use **logistic regression** model

The study was also limited to only the use of **synthetic (artificial) datasets**.

06. Conclusions

Broader Impact on XAI

This research found the important and needs for more data documentation standard in Responsible AI practices. Usually, AI fairness has focused on model transparency and explainability, but this research shows that documenting dataset composition is also very important.

By not addressing dataset homogeneity/ imbalance and how small sample sizes is, the biases can go undetected, and then leading to discriminative result in critical decision-making processes.

The study underlines the importance of moving beyond model explainability. Transparency at both the model and dataset levels is essential for building fair AI systems.

Recommendation for Future Research

1.Bias Mitigation Techniques:

develop bias mitigation techniques in more detail, for example, Synthetic Minority Over-sampling (SMOTE) or Under-sampling Technique.

2.Exploring Other Models

This study was performed using logistic regression; while future research should include similar analysis of more complex models.

3. Explore the Impact of other Dataset Characteristics:

investigate how the differences in the dataset's characteristics such as feature noise, multivariate normality, correlation between features etc.

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Thank you

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